CS & IT ENGINEERING



Operating System

Memory Management

DDP_01 Discussion Notes



By- Saheb Tung sir

[MCQ]

#Q. Consider a fixed partition contiguous memory management technique, where there are 5 partitions of size 100MB, 250MB, 200MB, 500MB and 300MB. All Partitions are initially empty. The following process requests are made in the given order:

Process	Size
P1	150 MB
P2	400 MB
P3	270MB
P4	180MB
P5	80MB

The following 2 answers are calculated for First fit, Best fit and Worst Fit policies

1. Maximum degree of multiprogramming?
2. What is the total internal fragmentation size?

For which of the following partition allocation policy the above two calculations are not exactly same as other 2 policies?

A Best fit

B First fit

C Worst fit

D None of the above

Ans C

Process → 150MB, 400MB, 270MB, 150MB, 80MB.
P_1 P_2 P_3 P_4 P_5

First Fit

MM

100 MB	P_5 20MB
250MB	P_1 100MB
200MB	P_4 20MB
500MB	P_2 100MB
300MB	P_3 30MB

Deg Multiprogramming = 5

Total IF = 270 MB.

Best fit

MM

100MB	P_4 20MB
250MB	P_1 70MB
200MB	P_1 50MB
500 MB	P_2 100MB
300 MB	P_3 30MB

Deg Multiprog = 5.

Total I.F = 270MB.

Worst Fit.

MM

100 MB	
250MB	P_4 70MB
200 MB	P_5 120MB
500 MB	P_1 350MB
300 MB	P_3 30MB.

Deg multi prog = 4

Total I.F = 570 MB.

[NAT]

#Q. Consider a paged memory system where the logical address is 25 bits and physical address is 33 bits. The page size is 4KB. The approximate size of page table size is __21__ k bytes ($1k = 2^{10}$)?

Page table size = No of pages × each page table entry size.

$= 2^{13} \times 21$ bits

$= 2^{10} \times 21 \times 8$ bits

$= 21$ KB

Ans 21

P.S = 4KB
$= 2^{12}$ B

L.A = 25 bit

page no bit.	offset
13 bits	12 bits

Total pages
$= 2^{13}$

P.A = 33 bit

frame no bits	offset
21 bits	12 bits

[NAT]

#Q. Consider a paged memory system where the page number is 12 bits and physical address is 33 bits. The page size is 2KB. The approximate size of page table size is ____11____k bytes ($1k = 2^{10}$)?

P.A = 33 bit

frame no. bits	offset
22 bits	11 bits

P.S = 2KB = 2^{11} B

No of Pages = 2^{12}

PT size = $2^{12} \times 22$ bits.

$= 11 \times 2^{13}$ bits.

$= 11 \times 2^{10} \times 8$ bits.

$= 11$ KB

Ans 11

[NAT]

#Q. Consider a paged memory system which has 4k pages each with 8kbytes size. The system maintains page table of 12k bytes. Each page table entry contains frame number, 1 valid bit, 1 modified bit and 1 replacement bit. The number of frames in the main memory is __2__ *2^{20}?

P.S = 8KB = 2^{13} B

no of Pages = 4K

PT Size = 12K bytes.

no of pages × each page table entry size = 12K × 8 bits.

4K × (f + 3) bits = 12K × 8 bits

f + 3 = 24 bits.

f = 21 bits.

No of possible frames in MM

= 2^{21}

= 2×2^{20}

Ans 2

[MCQ]

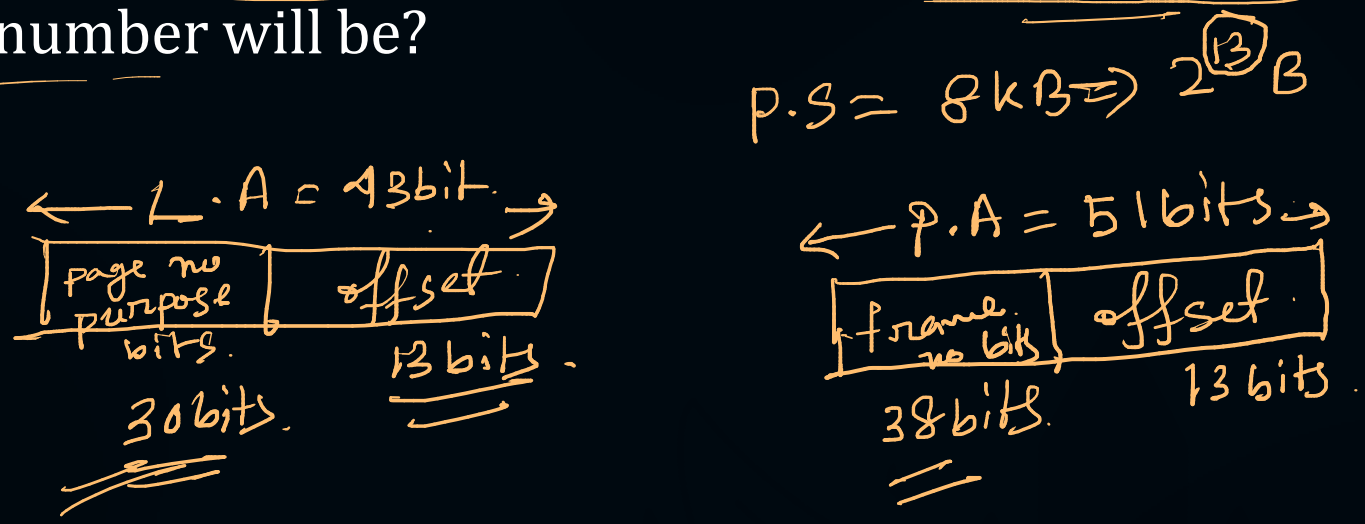
#Q. A system has 43-bit logical addresses and 51-bit physical addresses. If the pages are 8 kB in size, the number of bits required for logical page number and physical frame number will be?

A. 43 bits, 51 bits

B. 30 bits, 51 bits

C. 43 bits, 38 bits

D. 30 bits, 38 bits



Ans D

[NAT]

#Q. Consider a logical-address space of 8 pages, with page size 1024 bytes. The physical memory contains 32 frames. The page table size is 40 bits?

5 bit.

PT size = no of pages × entry size.

= 8 × 5 bits.

= 40 bits.

Ans 40

[NAT]

#Q. Consider a system using TLB for paging with TLB access time of 40ns. The hit ratio is to be used for TLB to reduce the effective memory access time from 400ns without using TLB to 280ns with using TLB, is 80 %??

$T_{TLB} = 40\ ns.$

$E.M.A.T_{without\ TLB} = 400\ ns.$

$2t_{mm} = 400\ ns.$

$t_{mm} = 200\ ns.$

$H_{TLB} = ?$

$EMAT_{TLB} = H_{TLB}(T_{TLB} + t_{mm}) + (1 - H_{TLB})(T_{TLB} + t_{mm} + t_{mm})$

(PT, Content)

$280 = H(40 + 200) + (1-H)(40 + 400)$

$280 = 240H + 440 - 440H$

$200H = 160$

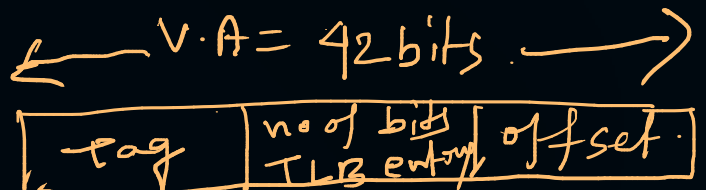
$H = 8\frac{160}{200} = .8$

Ans 80

[NAT]

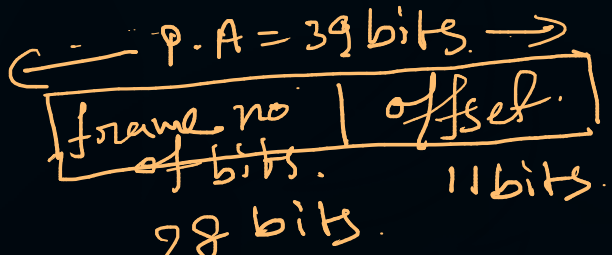
#Q. A computer system implements a 42- bit virtual address, 512GB physical address space, page size of 2KB, and an 8KB look-aside buffer (TLB) organized as direct mapped. Each page table entry contains a valid bit, a dirty bit and 2 protection bits along with the translation. The minimum length of the TLB tag in bits is 20 bits

P.S = 2KB = 2^{11} B | P.A.S = 512GB = 2^{39} B



PT each entry size = (f + 4) bits

= (28 + 4) = 32 bits = 4B



tag length = 20 bits

no of PT entry in TLB = $\frac{8KB}{4B} = 2K = 2^{11}$

Ans 20

THANK - YOU